

TAIMUR HASSAN SARMAD

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EDUCATION

Lahore University of Management Sciences

Bachelor of Science in Computer Science (Major **CGPA: 3.90/4.00**)

Sep. 2019 – June 2023

Lahore, Pakistan

WORK EXPERIENCE

FullStack Software Engineer

June 2023 – Present

PosterMyWall

TypeScript, React.js, Fabric.js, Redux, Storybook, PHP, SQL, AWS, Puppeteer

- Built a full-stack **multi-agent AI** system on an extended **NeuronAI** framework powering PosterMyWall's AI Chat, with **concurrent multi-chat streaming** via **Server-Sent Events** and a **React** client handling live token streams across parallel sessions; drove **165k+ chats** and **96k+ generated assets** at a **75% positive feedback rate** during a 6-week unmarketed soft launch.
- Added **parallelism** to the agent/node architecture and designed a hybrid **agentic multimodal context formation** layer (chosen over RAG-style embedding retrieval) that sends only relevant labeled chat assets to downstream agents, cutting workflow latency by **6–8s** and lifting positive feedback ratio by **70%**.
- Architected a feature enabling users to add AI-generated/manual subtitles to audio/video content, sync and edit them using a timeline UI made in **React**, and apply advanced styling/animations to them in a custom **Fabric.js** group.
- Scaled stock media offerings from 2,000 items in 2 categories to 300,000+ items across 8 categories by optimizing **MySQL** tables and relationships, automating optimization (using **ImageMagick** and **FFmpeg**) for presets stored on **AWS S3**, and implementing AI-based keyword/category tagging for efficient filtering and search with **AWS CloudSearch** and **OpenSearch**.
- Reverse-engineered the 2D transformation math behind **Fabric.js**'s resize, scale, and rotate controls, to rebuild them as HTML/CSS overlays that operate on canvas objects positioned outside the visible viewport too.
- Built a constraint-aware color assignment algorithm that applies and reshuffles palettes across multi-layer canvas designs in real time, analyzing **z-index** stacking and geometric overlaps to guarantee visual contrast between touching items.
- Refactored **57 React components**, removing duplicated state and anti-pattern **useEffects** in favor of **reusable hooks** and **memoized selectors**, eliminating unnecessary re-renders and improving readability.

PROJECTS

Lahore AQI Updates - Punjab Government EPA

Feb. 2025 – Present

TypeScript, React Native, Node.js, Express, MongoDB, InfluxDB, Firebase

- Developed a **React Native Android App** with **TypeScript**, delivering real-time and historical air quality data visualisations with interactive charts, 24-hour AQI forecasts, and personalized location-based push notifications.
- Handled state management using **React Context API**, custom hooks, AsyncStorage for persistent data with versioning, geolocation services, push notifications via **Firebase Cloud Messaging**, internationalization with **i18n**, device permissions management, and automated app updates.
- Built a scalable **Node.js + Express.js** backend that fetches the latest readings from 9 Lahore-wide EPA monitors, serves screen-specific data, and runs **cron** jobs to fetch daily weather information and run an ML model updating next-day AQI forecasts; using **Redis** for caching, semaphore locking, and session management.
- Integrated **MongoDB** to store user preferences, activity, and monitor JSON data; later migrated monitor data to **InfluxDB** when the government upgraded from hourly to minute-level readings, **reducing storage costs by 30x** and **optimising** historical time-range queries by **200x**.

FactCheck: A Multilingual Web Scraper for Resource-Constrained Devices

June 2023 – Dec. 2023

AWS, HTML, CSS, JS, React.js, Flask, Web-Scraping (beautifulSoup4), Cron, Caching, SSR

- Created an automated Python **web scraper** to fetch and translate fact-checked articles into Urdu, using a **cron** job to schedule Google Translate API calls and implemented **caching** with **Redis**, which reduced API costs by over 95%.
- Developed a web app (*no longer hosted*) to display the articles using React and Python (Flask), and optimized it for low-resource devices by implementing **Server-Side Rendering**, **lazy-loading**, and **infinite scroll**. Configured an **Apache** server on a **EC2** Ubuntu instance to deploy the webapp.

An Android Exploit for Audio Eavesdropping | React Native, Android Development

Jan 2023 – May 2023

- Developed an Android app with automated scripts to create Google Assistant-narrated audio files, record accelerometer data at customizable sampling frequencies, and convert the raw data into noise-free mel-spectrograms.
- Bypassed Android 12's 200 Hz sampling limit by exploiting sensor management vulnerabilities and developed a kernel-level interleaving **algorithm** to double IMU sampling frequency without user permissions.

TECHNICAL SKILLS

Languages and Frameworks: JavaScript/TypeScript, NodeJS, ReactJS, React Native, AngularJS, Express, C/C++, Python, PHP, CodeIgniter, SQL, HTML/HTML5, CSS, JQuery

Developer Tools: Git, SVN, AWS (S3, EC2, CloudSearch, DynamoDB), MongoDB, InfluxDB, Google Cloud Platform, axe DevTools, WAVE, Docker, Storybook, Redux, Webpack, Expo, Cron, Bash Scripts, Apache/Nginx Server Configurations